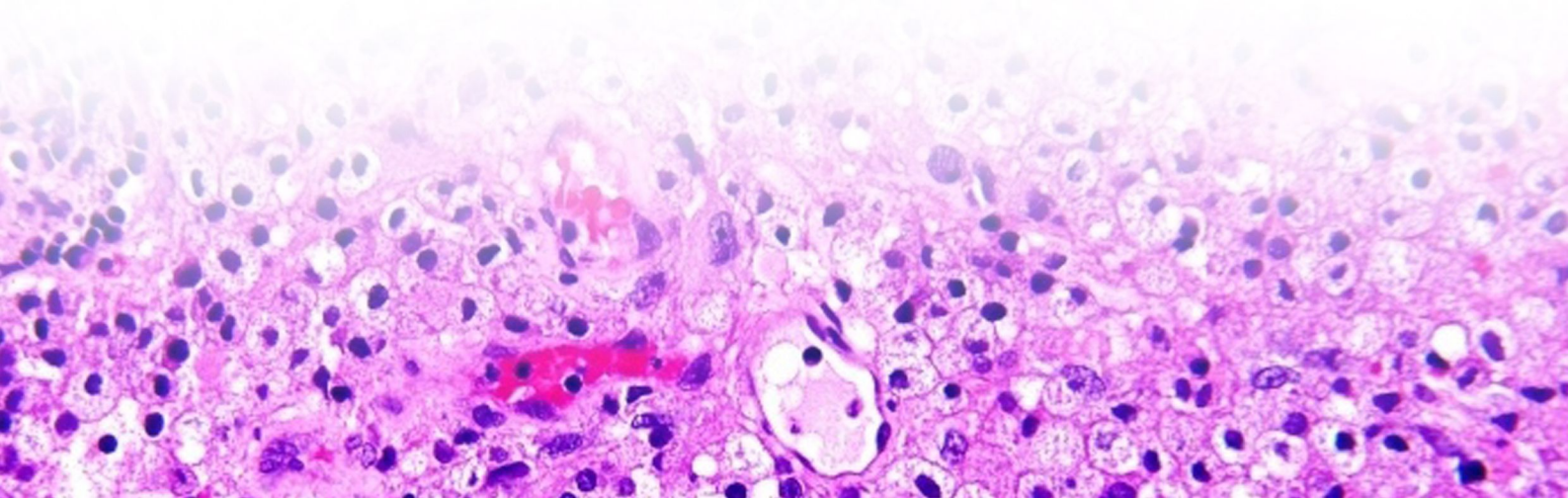
Student Notes

PATHOLOGY

1st Edition

Unit 15

Head and Neck



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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HEAD AND NECK

SHORT NOTES:

1. Salivary Adenoma
2. Paraganglioma
3. Warthin tumour
4. Thyroglossal cyst

SHORT ANSWERS:

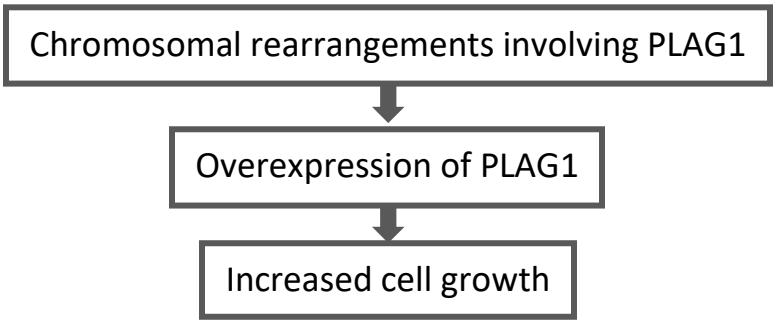
1. Nasopharyngeal CA
2. Malignant tumours of salivary gland
3. Etiological factors associated with squamous cell CA of oral cavity
4. Sites of oncocystoma
5. Ameloblastoma
6. Adamantinoma Jaw
7. Salivary Gland Tumors Classification

SHORT NOTES:

1. SALIVARY ADENOMA

- **Etiology:**
  - M/C benign tumor
  - Mixture of ductal and myoepithelial cells, show both epithelial and mesenchymal differentiation - **Mixed tumor**
  - Tumor neoplastic elements are of myoepithelial or ductal reserve cell origin - **Pleomorphic adenoma.**
- **Site:**
  - Parotid gland- most common
  - Submandibular and minor salivary glands- rare

PATHOGENESIS:



MORPHOLOGY:

- **Shape:** round
- **Size:** not more than 6 cm
- **Capsule:** encapsulated; not fully developed in palate; small protrusions into surrounding gland, if present, lead to recurrences after enucleation
- **Cut surface:** grey white, cartilage like

MICROSCOPY:

- **Epithelial component:**
  - Arranged in the form of ducts, acini, irregular tubules
  - Ducts lined by cuboidal to columnar cells, surrounded by small myoepithelial cells
  - May show strands/sheets of plasmacystoid/spindled myoepithelial cells
- **Mesenchyme like elements:**
  - Epithelial elements are dispersed within a varying amount of mesenchyme like background of loose myxoid tissue, islands of hyaline, chondroid and mucoid matrix

CLINICAL FEATURES: Painless, slow growing, mobile, discrete masses.

2. PARAGANGLIOMA

- Chromaffin cells+ extra adrenal neuroendocrine cells → PARAGANGION SYSTEM.
- Associated with sympathetic and parasympathetic nervous system.
- Tumors from paraganglia →paraganglioma.
- M/C paraganglioma → pheochromocytoma.
- M/C extraadrenal paraganglioma → at head and neck region.

SITE:

- Paravertebral paraganglia (organs of zuckerkandl, bladder)
- Paraganglia along great vessels (carotid bodies, aortic bodies, etc)

PATHOGENESIS:

- Sporadic/ familial
- Familial- autosomal dominant, along with MEN-2.
- Mutations in SDH gene/ genes participating in mitochondrial oxidative phosphorylation → alteration in cellular metabolism → slow growing painless masses.
- AGE: 50-60 years.

MORPHOLOGY: (carotid body tumor)

- **Site:** bifurcation of common carotid artery.
- **Number:** single.
- **Size:** not more than 6 cm.
- **Color:** red-pink → brown.
- **Microscopy:**
  - **Cellularity:** chief cells, sustentacular cells.
  - **Arrangement: zellballen pattern.**
  - **Cytoplasm:** abundant, clear/granular, eosinophilic.
  - **Nuclei:** uniform, round-ovoid, vesicular

STAINING/ IMMUNOREACTIVITY:

- Chief cells; chromogranin, synaptophysin, neurospecific enolase, CD56, CD57.
- **Sustentacular cells:** antibodies against S-100.
- **Electron microscopy:** well demarcated neuroendocrine granules.

3. WARTHIN TUMOUR (Papillary Cystadenoma Lymphomatosum)

- Second most common salivary gland neoplasm.
- Exclusively in parotid gland (the only tumor restricted to parotid)
- M>>F, 50-70yrs, 10% Multifocal & 10% Bilateral.
- Smokers have 8X the risk of non-smokers.
- **Morphology:** round to oval encapsulated masses, superficially palpable.
- Transection shows pale gray surface punctuated by narrow cystic/ cleft-like spaces filled with mucinous / serous secretions.
- **Microscopically,** double layered neoplastic epithelial cells resting on dense lymphoid stroma bearing germinal centers.
    - **Upper layer:** palisading columnar cells with abundant granular (due to presence of mitochondria called oncolytic), eosinophilic cytoplasm.
    - **Lower layer:** cuboidal/polygonal cells.
- Secretory cells may be seen. On occasion, they are foci of squamous metaplasia.
- Benign, recurrence rate only 2% after resection.

4. THYROGLOSSAL DUCT CYST

DEVELOPMENT OF THYROID GLAND:

Site: foramen cecum, at base of tongue → descends down → this pathway forms a temporary duct → thyroglossal duct → connects the gland to the base of tongue → disappears once the gland reaches its definitive site in the anterior neck.

DEVELOPMENT OF CYST:

If the duct does not disappear completely → cyst develops from the remnants → thyroglossal duct cyst.

MORPHOLOGY:

- 1-4 cm diameter
- **Lining:**
    - stratified squamous (near tongue base)
    - pseudo-stratified columnar (anywhere below)
    - Transitional pattern
- **Malignant transformation:** rare

TREATMENT: Surgical excision.

SHORT ANSWERS:

1. NASOPHARYNGEAL CARCINOMA

- Close anatomic relationship to lymphoid tissue
- Associated with **EBV infection**.
- Disease take any one of the three patterns:
  - Keratinizing squamous cell carcinoma.
  - Non-keratinizing squamous cell carcinoma.
  - Undifferentiated/basaloid carcinoma.
- Three factors influence the origin of neoplasms:
  - Heredity
  - Age
  - Infection with EBV.

2. MALIGNANT TUMORS OF SALIVARY GLAND

Mucoepidermoid carcinoma:

- Composed of variable mixtures of squamous cells, mucus secreting cells and intermediate cells.
- Most common site: parotid gland.
- (11;19) (q21; p13) chromosomal translocation.
- MECT1 – MAML2 fusion gene.
- Most common form of primary malignant tumor of salivary glands.

Adenoid cystic carcinoma:

- Uncommon tumor.
- Most common location among minor glands – palatine glands.
- Most common location among major gland – parotid and submandibular glands.

Acinic cell carcinoma:

- Resemble normal serous acinar cells of salivary glands.
- Most common site – parotid.
- Rarely involves minor glands.

3. ETIOLOGICAL FACTORS A/W SCC OF ORAL CAVITY

- Chronic abusers of smoked tobacco and alcohol
- Chewing of betel quid and paan
- Actinic radiation (sunlight) {pipe smoking}

4. SITES OF ONCOCYTOMA

- Epithelial neoplasm composed of large eosinophilic cells
- Arise from intercalated cells of collecting ducts
- Eosinophilic cells have numerous mitochondria
- **Gross appearance** :
    - Tumors are tan or mahogany brown
    - Well encapsulated

5. AMELOBLASTOMA

- Odontogenic epithelium & shows no ectomesenchymal differentiation.
- Commonly cystic, slow growing & locally invasive but had indolent course.
- Treatment: surgical resection.

6. ADAMANTINOMA JAW

- The most common form of ameloblastoma - the multicystic form - was formerly known as adamantinoma of the jaw.
- However, ameloblastoma is unrelated histologically to adamantinoma of the bone, and this terminology should be abandoned to avoid confusion.

7. SALIVARY GLAND TUMORS CLASSIFICATION:

BENIGN	MALIGNANT
Pleomorphic adenoma	Mucoepidermoid carcinoma
Warthin tumor	Adenocarcinoma
Oncocytoma	Acinic cell carcinoma
Basal cell adenoma	Adenoid cystic carcinoma
Canalicular adenoma	Malignant mixed tumor
Ductal papilloma	Squamous cell carcinoma

UPDATES FROM ROBBINS: 10th EDITION

1. SQUAMOUS CELL CARCINOMA OF HEAD AND NECK:

- Infection with high-risk human papillomavirus (HPV) is now the primary cause of SCC of the oropharynx.
- SCC of head and neck has been put into 2 groups now.
  - HPV associated SCC
  - Classic Non- HPV related SCC

- Both differ in several aspects.

ENTITY	HPV ASSOCIATED SCC	NON HPV RELATED SCC
MC Sites	Tonsillar crypts within the lingual tonsils, base of tongue, soft palate, and oropharynx.	Oral cavity- ventral surface of the tongue, floor of the mouth, lower lip, soft palate, and gingiva
Patient Age	Younger	Older
Risk factors	Oral sex	Tobacco, Alcohol
Clinical presentation	Small primary, bulky Lymph nodes swelling	Large primary, Variable lymph node involvement
Histology	Non keratinising SCC	Keratinising SCC
Distant metastasis	Rare	Common
Clinical outcome	Good	Poor
Risk of 2nd primary	Low	High
Specific mutations	P16 over expression, inactivation of p53 & RB.	More somatic mutations

2. MUTATIONS IN SCC:

- Driver mutations- TP53/ CDKN2A/ PIK3CA
- Other mutations- NOTCH1, FAT1 (tumor suppressor)

3. HISTOLOGY OF HPV-ASSOCIATED SCC:

- Proliferation of nests and lobules of non-keratinizing & basaloid cells growing within sheets of lymphocytes.
- Most cystic SCCs in the neck are metastases from cancers of the upper airways or digestive tract.

4. ODONTOGENIC CYSTS:

- The term PERIAPICAL GRANULOMA is replaced by RADICULAR CYST.
- Nasopalatine cyst is included as a developmental cyst

5. SINONASAL PAPILLOMA:

- 3 types
  - Exophytic,
  - Endophytic ,
  - Oncocytic (previously called cylindrical)
- Most endophytic sinonasal papillomas have EGFR gene mutations.
- The remaining cases generally harbor HPV DNA, often low-risk types 6 and 11.

6. ACINIC CELL CA:

- Composed of cells that resemble normal serous acinar cells of the salivary glands, with small and round nuclei and variable morphology
- Cytoplasm typically contains purple granules (zymogen granules recapitulating the normal serous acinar digestive enzymes), but cells can also be clear or vacuolated.
- The neoplastic cells are organized into sheets, microcysts, glands, follicles, or papillae.
- Sheet like growth pattern, nuclear pleomorphism, mitoses, and necrosis are referred to as de-differentiated or having undergone high-grade transformation.

CANCER	UPDATE
Sinonasal Papilloma	EGFR mutations, HPV 6 and 11
Nasopharyngeal Angiofibroma	CTNNB1 (β-catenin) mutations; Part of FAP
Paraganglioma (Familial)	Succinate Dehydeogenase B (SDHB) mutation are associated with highest rates of metastasis (⅓rd)
Pleomorphic Adenoma	Mutations of the HMGA2 gene, which encodes a DNA-binding protein, are associated with many of the cases that lack PLAG overexpression in PAs
Mucoepidermoid Carcinoma	Balanced (11;19) (q21;p13) chromosomal translocation that creates a fusion gene composed of portions of CRTC1 (MECT1) and MAML2 genes
Adenoid Cystic Carcinoma	MYB-NFIB gene rearrangements are present in a subset

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